

Saraswati Veena Vibro-Acoustic Protocol

Hackathon Fast-Track Matrix (Essential 180-Take Standard)

Vibro-Acoustic AI Research Team

August 2, 2026

1 Essential Setup & Execution Rules

- **15-Take Dynamic Standard:** Every state requires 15 takes divided into 3 plucking force tiers: 5 Soft (*Mridu*), 5 Medium (*Madhyama*), and 5 Hard (*Kathina*) to teach the AI dynamic force invariance.
- **Acoustic Decay Window:** Allow an uninterrupted 8-second decay per take without dampening.
- **Plucking Anchor Point:** Plucks are anchored at 14.11 cm from the main bridge (*Kudirai*) using a wire plectrum (*Nakhama*).
- **Sensor Placement:** Sensor module flat on top wooden faceplate (*Kudam Palakai*), 3–5 cm in front of the main bridge, with mic portal facing upward (10–15 mm clearance).

2 Master Fast-Track Data Collection Matrix (180 Takes Total)

Table 1: Fast-Track Data Collection Matrix for Saraswati Veena

Phase / Item	Carnatic Terminology	Execution Instructions	Repetitions Standard	AI Feature / Technical Rationale	Low-Cost Setup
Phase 1: Open Baselines	Sarani (S_1), Panchama (S_2), Mandra (S_3), Anumandra (S_4)	Pluck each open melody string individually with wire plectrum (<i>Nakhama</i>). Hold ring-out for 8 s.	60 takes total (15 per string: 5 Soft, 5 Med, 5 Hard)	Establishes baseline fundamental pitch (f_0), bridge down-force, and non-linear harmonic decay.	Natural instrument state.
Phase 2: Core Combo	Combo A ($S_1 + T_{1-3}$)	Record combined stroke (<i>Kootu Meettu</i>): Pluck S_1 and stroke side drones T_{1-3} simultaneously. Let ring for 8 s.	15 takes total (5 Soft, 5 Med, 5 Hard)	Captures multi-string bridge loading, intermodulation harmonics, and drone resonance coupling in 1 take.	Natural instrument state.
Phase 4: Body Impulse	Thattam / Naada Muting	Pluck S_1 sharply and immediately damp string vibration with palm within 0.1 s.	15 impulse takes (Immediate palm dampening)	CRITICAL: Isolates pure body Frequency Response Function ($H(f)$) and structural modal damping (ζ).	Manual hand dampening on string.
Phase 6: Fret Anchors	Frets 7 & 12 on S_1	Play single open strokes anchored at key frets: Fret 7 (Mid-Neck) and Fret 12 (Octave Center).	30 takes total (15 per fret: 5 Soft, 5 Med, 5 Hard)	Replaces full sweep: maps mid-neck stiffness uniformity and octave center nodal acoustic behavior.	Standard fret fingering.
Fault Label 01	Peg Slippage / Tension Shift (<i>Biridai</i>)	Pluck S_1 independently while intentionally detuned counter-clockwise by 15–20 Hz.	15 takes total (5 Soft, 5 Med, 5 Hard)	Teaches model f_0 pitch drift / tension loss without structural body degradation.	Peg detuning via tuner app.
Fault Label 02	Bridge Lip Buzz (<i>Javari Chatter</i>)	Execute Combo A while a thin paper shim (0.2 mm) is placed under brass bridge lip (<i>Kudirai</i>).	15 takes total (5 Soft, 5 Med, 5 Hard)	Induces high-frequency harmonic chatter and boundary reflection noise on spectrograms.	10 × 5 mm paper shim under bridge lip.
Fault Label 03	Shell Mass Damping (<i>Kudam Damping</i>)	Execute Combo A while 15 g mounting putty (Blu-Tack) is firmly affixed to outer shell (<i>Kudam</i>).	15 takes total (5 Soft, 5 Med, 5 Hard)	Alters coupled structural cavity modes and increases acoustic modal damping (ζ).	15 g Blu-Tack on resonator shell.
Fault Label 04	Fretbed Softening (<i>Melam Degradation</i>)	Pluck Fret 7 on S_1 with a rubber eraser slice wedged beneath the brass fret wire.	15 takes total (5 Soft, 5 Med, 5 Hard)	Models viscoelastic energy absorption under Fret 7 (simulates wax bed degradation).	Pencil eraser slice beneath fret wire.
Fault Label 10	Compound Fault (Tension + Buzz)	Pluck S_1 detuned by 15 Hz WITH paper shim inserted under brass bridge lip.	15 takes total (5 Soft, 5 Med, 5 Hard)	CRITICAL: Evaluates multi-label classification decision boundaries under dual mechanical defects.	Electronic tuner + paper shim.

3 Rationale for Backup Case Selection

This backup fast-track matrix reduces total recording time to **≈ 60 minutes** while preserving high information density for AI model training:

3.1 Healthy Baseline Selection

- **Phase 1 (Open Baselines):** Establishes ground-truth fundamental pitch (f_0) and tension baselines across all 4 melody strings (S_1 – S_4).
- **Phase 2 (Combo A):** Captures multi-string bridge down-force and drone sympathetic resonance in a single 15-take set, replacing isolated drone phases.
- **Phase 4 (Body Impulse / *Thattam*):** Quick palm-muting cuts string vibration, allowing the transducer to capture the pure wooden body response ($H(f)$) and modal damping (ζ).
- **Phase 6 (Frets 7 & 12):** Replaces a lengthy 24-fret sweep by targeting key structural nodes: Fret 7 (mid-neck flexure) and Fret 12 (octave midpoint/antinode).

3.2 Fault Case Selection

- **Fault 01 (Peg Slippage):** Teaches the AI to recognize pure pitch drop / tension loss (15–20 Hz) without physical body damage.
- **Fault 02 (Bridge Buzz / *Javari*):** Simulates non-linear bridge chatter, generating sharp high-frequency noise bursts on spectrograms.
- **Fault 03 (Shell Putty):** Dampens body vibration on the main bowl (*Kudam*), testing if the model can detect structural energy loss.
- **Fault 04 (Fretbed Softening):** Mimics wax bed (*Melam*) degradation under Fret 7, teaching localized neck-damping characteristics.
- **Fault 10 (Compound Fault):** Combines detuning + bridge buzz to test how well the AI handles multiple overlapping defects during a live demo.